

Introduction

A *data descriptor* (or data paper) publishes a dataset as a citable research object. Unlike an analysis paper, its contribution is not a statistical finding but a **contract**: a precise, machine-readable account of what the dataset contains, how it was produced, how it is licensed, and what quality guarantees it carries. The FAIR guiding principles — Findable, Accessible, Interoperable, Reusable (Wilkinson et al. 2016) — frame why this matters: a dataset is only reusable to the extent that its structure and provenance are legible to both humans and machines.

What this template provides I

This exemplar packages the recurring moving parts of a data descriptor so that a fork can start from a working, tested baseline:

What this template provides I

- ▶ A **schema / data dictionary**: named fields with declared types, nullability, units, and value constraints (patterns, enumerations, numeric bounds).
- ▶ A **file inventory**: each shipped file with its media type, a sha256 checksum, and a row count.
- ▶ A **provenance chain**: the ordered steps and agents that produced the release.
- ▶ A **license boundary**: an explicit reuse license, and the discipline of publishing *metadata and checksums* rather than bundling restricted bytes.
- ▶ A **validation gate**: a tested library that rejects malformed, unsafe, or incomplete descriptors and verifies declared checksums against real bytes.

The shipped dataset is **synthetic and deliberately small** — it exists to make the workflow runnable and testable, not to support any empirical claim. Its values are generated deterministically. Accordingly, this manuscript restricts its claims to dataset *structure, provenance, quality, and release readiness*. When a real dataset is forked in, the same validation and figure code applies unchanged; only the descriptor and fixture files are replaced. The sections that follow describe the demonstration dataset (sec. sec:data), its schema (sec. sec:schema), its provenance (sec. sec:provenance), and the quality gate that binds the descriptor to the bytes it names (sec. sec:quality), and close with usage and forking notes (sec. sec:usage).